Ch 01: Using Python

Application Program Development

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Using Python

- Python must be installed and configured prior to use
  - See Getting Started with Python and Git for Class Assignments
- Python interpreter can be used in two modes:
  - **Interactive mode**: enter statements on keyboard
  - **Script mode**: save statements in Python script (file) and pass to interpreter

Concept

The Python interpreter can run Python programs that are saved in files or interactively execute Python statements that are typed at the keyboard. Python comes with a program named IDLE that simplifies the process of writing, executing and testing programs.



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Interactive Mode

- When you start Python in interactive move, you will see a prompt
  - Indicates the interpreter is waiting for a Python statement to be typed
  - Prompt reappears after previous statement is executed
  - Error message displayed if you incorrectly type a statement
- Good way to learn new parts of Python
  - Or test out ideas when writing / designing applications

```
C:\Users\etamustudent> python3
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit
Type "help", "copyright", "credits" or "license" for more information.
>>> name = input('What is your name? ')
What is your name? Schrodinger
>>> print('Hello', name)
Hello Schrodinger
>>>
```



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Script Mode

- Statements entered in interactive mode are not saved as a program
- To have a program use script mode
  - Save a set of Python statements in a file
  - The filename should have the `.py` extension
  - To run the file, or script, type the following at the OS command line

```
C:\Users\etamustudent> python3 q02.py
What is your name? Schrodinger
Hello Schrodinger!
```

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The IDLE Programming Environment

- **IDLE** (Integrated Development Program): single program that provides tool to write, execute and test a program (Appendix B)
  - Automatically installed when Python language is installed
  - Runs in interactive mode
  - Has built-in text editor with features designed to help write Python programs
  - A basic IDE, but recommend using VSCode, PyCharm

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Class Resources Repository

The class resources for these videos, slides and other materials can be found here:

Class Resources Repository for CSci 233 Applications Program Development



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Summary

- You must install a Python distribution (V3)
- Python interpreter can be used in two modes
  1. Interactive mode: try things out and learn
  2. Script mode: save statements in a file and run
- The IDLE programming environment is available to use
  - Though consider VSCode or PyCharm

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References

